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CODE:

```
# Implementation of Logistic Regression Algorithm

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix

# Load the Iris Dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# Create Logistic Regression Model
model = LogisticRegression(max_iter=200)

# Train the Model
model.fit(X_train, y_train)

# Predict the Test Data
y_pred = model.predict(X_test)

# Calculate Accuracy
accuracy = accuracy_score(y_test, y_pred)

# Calculate Confusion Matrix
cm = confusion_matrix(y_test, y_pred)

# Display Results
print("LOGISTIC REGRESSION RESULTS")
print("---------------------------")

print("\nAccuracy:", round(accuracy * 100, 2), "%")

print("\nConfusion Matrix:")
print(cm)

# Predict a New Sample
new_sample = [[5.1, 3.5, 1.4, 0.2]]

prediction = model.predict(new_sample)

print("\nNew Sample:", new_sample[0])
print("Predicted Class:", iris.target_names[prediction[0]])
```

OUTPUT :

```
LOGISTIC REGRESSION RESULTS
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Accuracy: 100.0 %

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

New Sample: [5.1, 3.5, 1.4, 0.2]
Predicted Class: setosa
```